

Q-1

Find the moments of inertia of a ring of radius $2a$ about an axis through center?

Sol:-

m-m

Let M be the mass and $2a$ radius of the ring. Then the mass per unit length will be $M/2\pi 2a$. We regard this ring to be composed of small element of mass (δm) each of length δs , can be written as

$$\frac{\delta m}{\delta s} = \frac{M}{2\pi 2a} \Rightarrow \delta m = \frac{M}{2\pi 2a} \delta s$$

Moment of inertia of the element about an axis through center O and perpendicular to the plane of the ring equals $\frac{M}{2\pi 2a} \delta s (2a)^2$. Therefore, moment of inertia of whole ring will be

$$\begin{aligned} I_{\text{ring}} &= \frac{M}{2\pi 2a} (2a)^2 \sum_{\text{element}} \delta s = \frac{4Ma^2}{4\pi a} \int ds = \frac{Ma}{\pi} \int ds \\ &= \frac{Ma}{\pi} \times 2\pi 2a \\ &= 4Ma^2 \end{aligned}$$

Hence $4Ma^2$ is a required moment of inertia

Q.2

Find the moment of inertia of a ring of radius a about an axis through center?

Sol:-

m-m

Let M be the mass and a is the radius of the ring. Then the mass per unit length will be $\frac{M}{2\pi a}$. We regard this ring to be composed of small element of mass (δm) each of length δs , can be written as

$$\frac{\delta m}{\delta s} = \frac{M}{2\pi a} \Rightarrow \delta m = \frac{M}{2\pi a} \delta s$$

Moment of inertia of the element about an axis through center O and the perpendicular to the plane of the ring equals $\frac{M}{2\pi a} \delta s a^2$.

Therefore, the moment of inertia of whole ring,

$$\begin{aligned} I_{\text{ring}} &= \frac{M}{2\pi a} a^2 \sum_{\text{element}} \delta s = \frac{M a^2}{2\pi a} \int ds = \frac{M a}{2\pi} \int ds \\ &= \frac{M a}{2\pi} \times 2\pi a \\ &= M a^2 \end{aligned}$$

Hence,

$M a^2$ is the required moment of inertia.

Q.3

Calculate the moment of inertia of a uniform rod of length l about an axis $\perp$ to the rod and passing through an end point?

Sol:-

m.m

(Let x -axis be chosen along the length of the rod, with origin at one end point. Let M and a be the mass and length of rod respectively and suppose the rod to be composed of small elements.) $\rightarrow$ Note: یہ سوال میں دیا گیا ہے

Let dm is mass and dx be the length of specific element of rod at distance x from ~~end point~~ ^{end point} O .

$$\text{then, } \frac{dm}{dx} = \frac{M}{a} \Rightarrow dm = \frac{M}{a} dx$$

M.I of the element is

$$I_{\text{element}} = dm x^2 = \frac{M}{a} x^2 dx$$

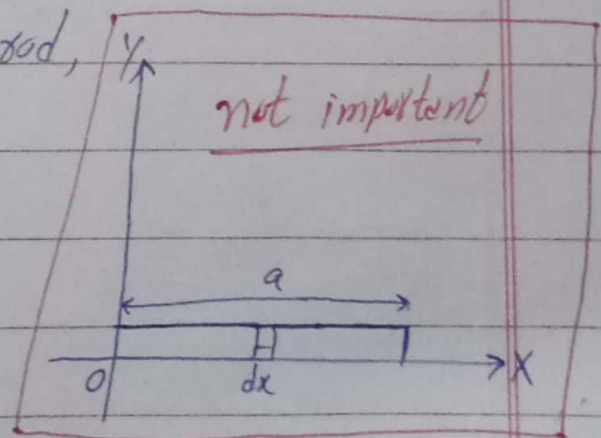
Hence, M-I of the whole rod,

$$I = \sum_{\text{element}} \frac{M}{a} x^2 dx$$

$$= \frac{M}{a} \int_0^a x^2 dx$$

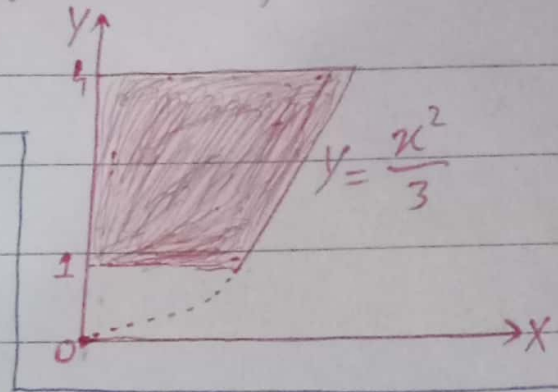
$$= \frac{M}{a} \left[\frac{x^3}{3} \right]_0^a$$

$$= \frac{M}{a} \cdot \frac{a^3}{3} = \frac{1}{3} M a^2$$



Q.4

Calculate the moment of inertia of the shaded area given in fig.



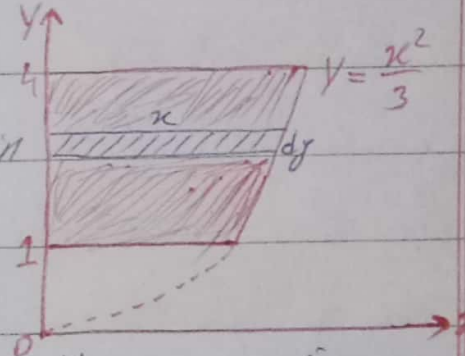
Sol: - m.m

We have, $y = \frac{x^2}{3} \rightarrow (i)$

We consider an elementary strip from the shaded area whose M-I is,

$$I_{\text{strip}} = \frac{1}{3} x^2 (x dy) = \frac{1}{3} x^3 dy$$

Then by integration, we obtain the moment of inertia of whole shaded area,



$$I_y = \sqrt{3} \int_1^4 y^{3/2} dy = \sqrt{3} \left| \frac{y^{3/2+1}}{\frac{3}{2}+1} \right|_1^4 = \sqrt{3} \left| \frac{y^{5/2}}{\frac{5}{2}} \right|_1^4$$

$$= \sqrt{3} \cdot \frac{2}{5} \left| y^{5/2} \right|_1^4 = \frac{2\sqrt{3}}{5} \left((4)^{5/2} - (1)^{5/2} \right)$$

by (i)

$$y = \frac{x^2}{3} \Rightarrow xy = x^2$$

$$\sqrt{3}y = x$$

$$\text{integrate w.r.t } y$$

$$\int \sqrt{3}y dy = \int x dy$$

$$\sqrt{3} \int y^{1/2} dy = 0$$

$$\sqrt{3} \cdot \frac{y^{3/2}}{\frac{3}{2}} = 0$$

$$\sqrt{3} y^{3/2} = 0$$

$$= \frac{2\sqrt{3}}{5} \left(((2)^2)^{5/2} - 1 \right)$$

$$= \frac{2\sqrt{3}}{5} \left((2)^5 - 1 \right)$$

$$= \frac{2\sqrt{3}}{5} (32 - 1)$$

$$= \frac{2\sqrt{3}}{5} (31) = \frac{62\sqrt{3}}{5}$$

$$= 21.5 \text{ inch}^4$$

Q.5

Consider a small element of uniform elliptic plate has moment of inertia $I = \delta m y^2$

Write down the expression for moment of inertia of whole plate? m-m
 (Note: Do not solve the expression)

Sol:-

The moment of inertia of whole plate will be

$$I_x = \int (\delta m) y^2$$

$$= \int_{\text{Plate}} \rho ds y^2$$

here $\rho = \frac{M}{\pi ab}$

Answer.

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Q.6

What is the impulse developed by a mass of 100gm which changes its velocity from $i+j+k$ to $2i+2j+3k$ m/sec?

Sol:-

$$\text{Mass} = m = 100 \text{ gm}$$

m-m

$$V_1 = i+j+k$$

$$V_2 = 2i+2j+3k$$

We know impulse,

$$\text{impulse} = P_2 - P_1 \quad \therefore P = mv$$

$$= mV_2 - mV_1$$

$$= m(V_2 - V_1)$$

$$= 100(2i+2j+3k - (i+j+k))$$

$$= 100(2i+2j+3k - i - j - k)$$

$$= 100(i+j+2k) \text{ Answer.}$$

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Q.7

A coordinate system rotates with angular velocity $\vec{\omega} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ relative to the fixed coordinate system $\vec{A} = \sin t \hat{i} - \cos t \hat{j} + e^{-t} \hat{k}$ then find $\frac{d\vec{A}}{dt}$ w.r.t fixed ^{point} system. when

$$\left(\frac{d\vec{A}}{dt}\right)_r = -\sin t \hat{i} + \cos t \hat{j} - 2e^{-t} \hat{k} \text{ and}$$

$$\vec{\omega} \times \vec{A} = (4e^{-2t} - 3\cos t)\hat{i} - (-2e^t + 3\sin t)\hat{j} + (\cos t + 2\sin t)\hat{k}$$

Soln: - m.m

To find $\frac{d\vec{A}}{dt}$ in the fixed system. we apply formula

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \vec{\omega} \times \vec{A} \rightarrow \textcircled{1}$$

All the values are given in question. put in $\textcircled{1}$

$$\left(\frac{d\vec{A}}{dt}\right)_f = (-\sin t \hat{i} + \cos t \hat{j} - 2e^{-t} \hat{k}) + (4e^{-2t} - 3\cos t)\hat{i} - (-2e^t + 3\sin t)\hat{j} + (\cos t + 2\sin t)\hat{k}$$

$$= (4e^{-2t} - \sin t - 3\cos t)\hat{i} + (3\sin t - 2e^{-t} + \cos t)\hat{j} + (-2e^{-t} + \cos t + 2\sin t)\hat{k}$$

Ans.

Q.8

To find the inertia matrix of uniform solid cuboid (parallelepiped) at one of its corner, if $I_{xx} = \frac{M}{3}(b^2 + c^2)$ then find I_{yy} and I_{zz} ?

Sol:-

by definition,

نوٹ: اگر کسی چیز کا ہوگا تو مکمل ہوگا۔
اور نہ ہیچے جو رہا لیا ہے وہ نہ ہوگا۔

$$I_{11} = I_{xx}$$

$$= \int \rho(x)(y^2 + z^2) dv$$

m.m

since, it is a uniform solid cuboid, and density ρ is constant, Therefore,

$$I_{xx} = \rho \int_0^a \int_0^b \int_0^c (y^2 + z^2) du dy dz = \rho \int_0^a \int_0^b \int_0^c (y^2 + z^2) dy dz du$$

$$= \rho a \int_0^b \int_0^c (y^2 + z^2) dy dz$$

$$= \rho a \left[\int_0^b \int_0^c y^2 dy dz + \int_0^b \int_0^c z^2 dy dz \right] = \rho a \left(c \frac{b^3}{3} + b \frac{c^3}{3} \right)$$

$$I_{xx} = \rho \frac{abc}{3} (b^2 + c^2) \quad \therefore \rho = \frac{M}{abc}$$

$$= \frac{M}{abc} \cdot \frac{abc}{3} (b^2 + c^2) = \frac{M}{3} (b^2 + c^2)$$

Same as,

$$I_{yy} = \frac{M}{3} (a^2 + c^2)$$

and

$$I_{zz} = \frac{M}{3} (a^2 + b^2)$$

2 چیز کے لیے ہے نہ کہ 1

by definition, $I_{11} = I_{xx}$

Given, $I_{xx} = \frac{M}{3} (b^2 + c^2)$

Therefore, $I_{yy} = \frac{M}{3} (a^2 + c^2)$

Q.9

If moment of inertia of a uniform circular cylinder of length b and radius a about x -axis through the center and $\perp$ to the central axis, is

$$I_x = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

sol:-

m.m

by Parallel-axis theorem, M-I of thin disc about x -axis is,

$$dI_x = \frac{1}{4}a^2 dm + z^2 dm = \left(\frac{1}{4}a^2 + z^2\right) dm$$

here $dm = \rho \pi a^2 dz$, thus.

$$I_x = \int_{-b/2}^{b/2} \left(\frac{1}{4}a^2 + z^2\right) dm = \int_{-b/2}^{b/2} \left(\frac{1}{4}a^2 + z^2\right) \rho \pi a^2 dz$$

$$= \rho \pi a^2 \int_{-b/2}^{b/2} \left(\frac{1}{4}a^2 + z^2\right) dz$$

$$= \rho \pi a^2 \left[\frac{1}{4}a^2 z + \frac{1}{3}z^3 \right]_{-b/2}^{b/2}$$

$$= \rho \pi a^2 \left(\frac{1}{4}a^2 b + \frac{1}{12}b^3 \right)$$

$$= \rho \pi a^2 b \left(\frac{1}{4}a^2 + \frac{1}{12}b^2 \right)$$

The mass of cylinder is $m = \rho \pi a^2 b$

thus

$$I_x = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right)$$

Due to symmetry we have

$$I_x = I_y = m\left(\frac{1}{4}a^2 + \frac{1}{12}b^2\right) \quad \text{Ans}$$

المoment

Q.10

Write the expression for total energy of a particle executing S.H.M.?

Answer:-

The total energy of a particle executing simple Harmonic Motion (S.H.M) ~~with~~ is,

Total energy = Kinetic energy + Potential energy

$$E = K.E + P.E$$

$$= \frac{1}{2}mv^2 + \frac{1}{2}Kx^2$$

m.m

Q.11

The moment of inertia of a rectangular plate with side a and b about x-axis

$I_x = \frac{1}{3}Mb^2$ and y-axis $I_y = \frac{1}{3}Ma^2$. Find moment of inertia along z-axis?

Sol:-

We know,

$$I_z = I_x + I_y$$

$$= \frac{1}{3}Mb^2 + \frac{1}{3}Ma^2$$

$$= \frac{1}{3}M(a^2 + b^2).$$

Answer.

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Q.12

What is a degree of freedom of the system of rigid bodies?

Ans:-

m.m

The number of coordinates required to specify the position of a system of one or more particles is called a degree of freedom of the system.

Example:-

A particle moving freely in space requires 3 coordinates. e.g. (x, y, z)

Q.13

Write down the mathematical expression for angular momentum of single particle?

Ans:-

Angular Momentum:

The angular momentum of a single particle is defined as the cross product of linear momentum and ~~positive~~ position vector of concerned particle.

Mathematically

$$L = r \times mv$$

Q-14

Two objects push off against one another starting from a stationary position. First object with mass 65 kg acquires a velocity of 2 m/s to the right. What velocity does the 2nd object having mass 70 kg acquire?

Soln:-

(Look u_1, u_2)

Let u_1 and u_2 be the initial velocity of mass 65 kg and 70 kg respectively, which are both equal to zero, because both objects are in stationary position. Let v_1 and v_2 be the final velocities of objects.

According to Law of conservation of linear momentum.

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$

since, $u_1, u_2 = 0$

$$m_1 v_1 + m_2 v_2 = 0 \rightarrow (1)$$

by (1) $v_2 = -\frac{m_1 v_1}{m_2} \rightarrow (2)$

Given $m_1 = 65 \text{ kg}$, $v_1 = 2 \text{ m/s}$, $m_2 = 70 \text{ kg}$

$$v_2 = \frac{-(65 \text{ kg})(2 \text{ m/s})}{70 \text{ kg}} = -\frac{130}{70} \text{ m/s} = -\frac{13}{7} \text{ m/s} = -1.8 \text{ m/s}$$

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Q.15

State Euler's theorem?

Ans:-

Theorem statement: m-m

A rotation of rigid body about a fixed point of body is ~~is~~ equal to a rotation about a line which passes through the point.

Q.16

State Chasles' Theorem?

Ans:-

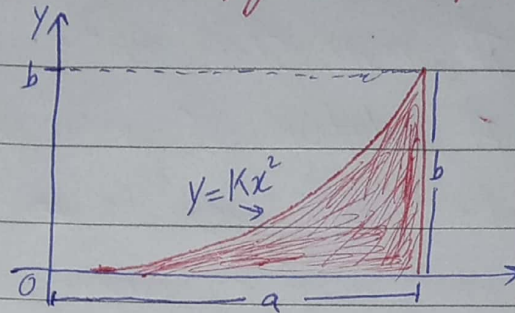
Theorem statement:

This theorem states that the most general rigid body displacement can be produced by a translation along a line (called its screw axis) followed by a rotation about that line.

Q.17

Write down the expression for moment of inertia of the shaded area shown in fig. with respect to x .

m-m →



Sol:-

Here we have $y = Kx^2 \rightarrow ①$

from fig. $x = a, y = b$ Put in ①

$$b = Ka^2 \Rightarrow K = \frac{b}{a^2}$$

Put the value of K in ①

$$\Rightarrow y = \frac{b}{a^2} x^2 \text{ or } x = \frac{a}{\sqrt{b}} \sqrt{y}$$

M-I along x -axis will be

$$\begin{aligned} I_x &= \int_A y^2 dA = \int_0^b y^2 (a - x) dy = \int_0^b y^2 \left(a - \frac{a}{\sqrt{b}} \sqrt{y} \right) dy \\ &= a \int_0^b y^2 dy - \frac{a}{\sqrt{b}} \int_0^b y^2 \cdot y^{1/2} dy = a \int_0^b y^2 dy - \frac{a}{\sqrt{b}} \int_0^b y^{5/2} dy \\ &= a \left[\frac{y^3}{3} \right]_0^b - \frac{a}{\sqrt{b}} \left[\frac{2}{7} y^{7/2} \right]_0^b \\ &= \frac{ab^3}{3} - \frac{2a(b)^{7/2}}{\sqrt{b}} \\ &= \frac{ab^3}{3} - \frac{2a(b)^{7/2-1/2}}{1} \\ &= \frac{ab^3}{3} - \frac{2ab^3}{2} = \frac{7ab^3 - 6ab^3}{21} = \frac{ab^3}{21} \text{ Ans.} \end{aligned}$$

Q-18

If $e_p = \cos\varphi \hat{i} + \sin\varphi \hat{j}$ and $e_\varphi = -\sin\varphi \hat{i} + \cos\varphi \hat{j}$ then prove that $\frac{de_p}{dt} = \dot{\varphi} e_\varphi$ and $\frac{de_\varphi}{dt} = -\dot{\varphi} e_p$ where dots denote differentiated with respect to t ?

sol:-

We have,

m.m

$$e_p = \cos\varphi \hat{i} + \sin\varphi \hat{j}$$

$$e_\varphi = -\sin\varphi \hat{i} + \cos\varphi \hat{j}$$

Then

$$\begin{aligned} \frac{de_p}{dt} &= \frac{d}{dt} (\cos\varphi \hat{i} + \sin\varphi \hat{j}) \\ &= -\sin\varphi \frac{d\varphi}{dt} \hat{i} + \cos\varphi \frac{d\varphi}{dt} \hat{j} \\ &= -\sin\varphi \dot{\varphi} \hat{i} + \cos\varphi \dot{\varphi} \hat{j} = \dot{\varphi} (-\sin\varphi \hat{i} + \cos\varphi \hat{j}) \\ &= \dot{\varphi} e_\varphi \end{aligned}$$

and

$$\begin{aligned} \frac{de_\varphi}{dt} &= \frac{d}{dt} (-\sin\varphi \hat{i} + \cos\varphi \hat{j}) \\ &= -\cos\varphi \cdot \frac{d\varphi}{dt} \hat{i} - \sin\varphi \frac{d\varphi}{dt} \hat{j} \\ &= -\cos\varphi \dot{\varphi} \hat{i} - \sin\varphi \dot{\varphi} \hat{j} \\ &= -\dot{\varphi} (\cos\varphi \hat{i} + \sin\varphi \hat{j}) \\ &= -\dot{\varphi} e_p \end{aligned}$$

Answer.

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Q.19

Write Necessary and sufficient condition for conservative force field?

m.m

Ans:-

Conservative force fields conserve the following properties.
 $\therefore$ iff = if and only if

i) A force field F is conservative iff there exists a continuously differentiable scalar field V such that $\vec{F} = -\nabla V$

or equivalently, iff $\text{curl } \vec{F} = \nabla \times \vec{F} = 0$

ii) A continuously differentiable force field F is conservative iff for any closed non-intersecting curve C (simple closed curve)

$$W = \oint_C \vec{F} \cdot d\vec{r} = 0$$

Q. 20

If the surface area of the ring is $2\pi r dr$ and the surface area of annulus is $\pi(R_2^2 - R_1^2)$

Then find the mass dm where mass of disk is M ?

Ans:-

 $m \cdot m$

The surface area of ring is

$$2\pi r dr$$

The surface area of annulus is

$$\pi(R_2^2 - R_1^2)$$

Therefore, we can have,

$$\frac{dm}{M} = \frac{2\pi r dr}{\pi(R_2^2 - R_1^2)}$$

$$dm = \frac{2r dr}{(R_2^2 - R_1^2)} \cdot M$$

Q.21

If $\mathbf{r} = a \cos \omega t \hat{i} + b \sin \omega t \hat{j}$ and $\mathbf{p} = -am\omega \sin \omega t \hat{i} + bm\omega \cos \omega t \hat{j}$, then find angular momentum?

Sol:-

m.m

Since angular momentum,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a \cos \omega t & b \sin \omega t & 0 \\ -am\omega \sin \omega t & bm\omega \cos \omega t & 0 \end{vmatrix}$$

$$= [abm\omega \cos^2 \omega t + abm\omega \sin^2 \omega t] \hat{k}$$

$$= abm\omega [\cos^2 \omega t + \sin^2 \omega t] \hat{k}$$

$$= abm\omega (1) \hat{k}$$

$$= abm\omega \hat{k}$$

is a required angular momentum.

Q.22

Find the M-I of solid circular cylinder of radius a , mass M and height of cylinder h about the axis of the cylinder, where M-I of one shell is,

$$I_{\text{shell}} = r^2 \frac{2M}{a^2} r dr = \frac{2M}{a^2} r^3 dr$$

Total M-I of solid circular cylinder is,

$$I_{\text{cylinder}} = \int_{r=0}^a I_{\text{shell}}$$

m-m

$$= \int_0^a \frac{2M}{a^2} r^3 dr$$

$$= \frac{2M}{a^2} \int_0^a r^3 dr$$

$$= \frac{2M}{a^2} \left[\frac{r^4}{4} \right]_0^a$$

$$= \frac{2M}{a^2} \left(\frac{a^4}{4} - 0 \right)$$

$$= \frac{Ma^2}{2} \text{ Ans.}$$

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Q.23

Use the parallel axis theorem to find the M-I of solid circular cylinder about a line on the surface of the cylinder and parallel to axis of cylinder?

sol:-

m-m

by parallel axis theorem.

$$I_A = I_c + Ma^2 \rightarrow \textcircled{1}$$

since I_c is the M-I of solid cylinder which is $I_c = \frac{1}{2} Ma^2$

by $\textcircled{1}$

$$I_A = \frac{1}{2} Ma^2 + Ma^2$$

$$= Ma^2 \left(\frac{1}{2} + 1 \right)$$

$$= \frac{3}{2} Ma^2$$

Answer.

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Q-24

Write down the expression for M-I of uniform hemisphere of radius a about its axis of symmetry, where $dv = dr(r d\theta)(r \sin\theta d\phi) = r^2 \sin\theta dr d\theta d\phi$.

Note: only write expression.

Ans:-

m-m

$$I_{zz} = \rho \int_0^a \int_0^{\pi/2} \int_0^{2\pi} r^4 \sin^3\theta dr d\theta d\phi$$

$$= \frac{2}{5} Ma^2$$

Ans.